

ENONTEKIO NAKKALA, Finland

WMO: 027260

Lat: 68.6031N

Lon: 23.5761E

Elev: 1227

StdP: 14.06

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.4	(c) -18.4	(d) -29.3	(e) 1.1	(f) -23.3	(g) -23.6	(h) 1.6	(i) -18.2	(j) 25.3	(k) 21.6	(l) 23.2	(m) 20.5	(n) 4.9	(o) 0	(p) 0.792

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 15.0	(c) 73.8	(d) 59.7	(e) 69.6	(f) 57.7	(g) 65.5	(h) 56.1	(i) 61.9	(j) 70.1	(k) 59.7	(l) 66.7	(m) 57.6	(n) 63.4	(o) 7.7	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.0	78.0	63.7	56.6	71.5	61.7	54.8	66.8	60.4	28.3	70.4	26.7	67.3	25.3	63.4	70.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 21.1	(b) 18.6	(c) 16.4	(e) -31.2	(f) 76.7	(g) 6.3	(h) 3.9	(i) -35.7	(j) 79.4	(k) -39.4	(l) 81.7	(m) -42.9	(n) 83.9	(o) -47.5	(p) 86.7		
(a) 21.1	(b) 18.6	(c) 16.4	(e) -31.2	(f) 64.0	(g) 6.0	(h) 3.4	(i) -35.5	(j) 66.5	(k) -38.9	(l) 68.4	(m) -42.3	(n) 70.3	(o) -46.6	(p) 72.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.6	7.1	8.2	15.1	26.2	39.1	48.5	55.5	51.4	43.8	29.2	17.7	12.7
	DBStd	19.33	12.41	13.73	10.68	8.13	8.29	7.21	6.71	6.03	5.84	8.19	10.86	11.85
	HDD50	7815	1331	1171	1082	715	354	113	21	56	199	646	970	1157
	HDD65	12914	1796	1591	1547	1165	803	496	302	424	637	1111	1420	1622
	CDD50	385	0	0	0	0	15	67	192	99	12	0	0	0
	CDD65	11	0	0	0	0	0	0	9	2	0	0	0	0
	CDH74	103	0	0	0	0	7	8	78	10	0	0	0	0
	CDH80	7	0	0	0	0	0	0	7	0	0	0	0	0
Wind	WSAvg	7.7	7.6	7.8	8.4	7.7	8.3	8.7	7.8	7.2	7.7	7.0	6.8	7.5
Precipitation	PrecAvg	18.5	1.0	0.9	0.6	0.8	1.4	2.2	3.0	2.5	1.9	1.7	1.3	1.1
	PrecMax	25.8	2.6	1.3	1.0	1.4	3.1	3.9	5.2	5.3	3.5	4.6	2.2	2.0
	PrecMin	13.9	0.2	0.3	0.3	0.2	0.3	0.7	1.0	1.1	0.6	0.5	0.4	0.2
	PrecStd	3.5	0.5	0.3	0.2	0.3	0.8	0.9	1.1	1.3	0.8	0.9	0.6	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	37.2	37.6	49.0	72.6	74.9	80.5	74.8	61.3	48.3	39.6	36.7
		MCWB	32.3	34.9	33.8	41.3	54.3	58.7	63.4	62.2	53.3	45.1	39.0	34.4
	2%	DB	29.4	32.1	34.5	43.0	63.7	69.1	75.8	69.7	57.8	44.9	34.8	32.8
		MCWB	28.5	30.7	31.7	36.9	50.5	55.5	60.9	58.4	52.0	43.1	33.9	32.0
	5%	DB	26.0	29.3	32.1	40.0	58.2	64.7	71.9	64.8	55.1	42.0	32.3	30.6
		MCWB	25.2	28.4	30.0	35.4	47.9	53.0	59.4	56.7	50.7	40.6	32.0	29.9
	10%	DB	23.0	26.5	29.8	37.4	52.3	60.2	67.4	61.3	52.5	39.3	30.8	28.7
		MCWB	22.4	25.7	28.1	33.8	44.5	51.4	57.9	55.2	49.0	38.1	30.5	28.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.6	35.5	34.1	41.6	55.5	62.8	65.3	65.2	56.3	45.7	39.4	34.8
		MCDB	34.0	37.5	36.2	48.4	70.5	72.3	73.5	70.4	58.7	47.1	39.5	36.4
	2%	WB	28.6	30.9	32.2	37.8	52.1	58.5	63.0	60.4	53.4	43.4	33.9	32.1
		MCDB	29.3	32.0	33.9	41.9	62.1	64.5	72.1	66.1	56.8	44.9	34.7	32.7
	5%	WB	25.3	28.5	30.4	35.8	48.1	55.6	60.9	57.9	51.5	40.6	32.1	30.1
		MCDB	26.0	29.2	31.9	39.6	57.8	61.5	68.9	63.0	54.2	41.9	32.3	30.5
	10%	WB	22.5	25.8	28.3	34.2	44.9	52.7	59.1	55.9	49.8	38.4	30.6	28.2
		MCDB	23.0	26.6	29.8	37.0	51.7	59.5	66.2	60.0	52.0	39.4	30.8	28.8
Mean Daily Temperature Range	5% DB	MDBR	14.5	14.4	16.9	15.0	14.3	14.3	15.0	14.5	11.8	9.7	12.1	13.8
		MCDBR	13.9	13.3	12.5	15.4	23.7	21.3	21.9	19.8	14.7	9.9	8.3	9.3
		MCWBR	13.6	12.5	10.6	10.6	12.2	10.4	9.1	9.9	9.3	7.9	8.2	9.0
	5% WB	MCDBR	13.5	12.9	11.9	13.6	22.4	17.6	18.2	17.1	12.0	9.1	8.2	8.9
		MCWBR	13.3	12.2	10.3	9.5	12.0	9.4	8.7	9.4	8.2	7.6	8.2	8.7
Clear-Sky Solar Irradiance	taub	0.180	0.236	0.276	0.305	0.324	0.329	0.336	0.321	0.290	0.251	0.182	N/A	
	taud	1.916	2.086	2.188	2.217	2.391	2.493	2.486	2.508	2.560	2.286	1.966	N/A	
	Ebn at Noon	24	173	239	266	275	277	271	262	244	171	21	0	
	Edh at Noon	2	14	24	32	30	29	28	24	17	11	2	0	
All-Sky Solar Radiation	RadAvg	8	143	618	1252	1602	1558	1462	1035	554	179	25	0	
	RadStd	1	11	27	56	154	112	142	73	56	27	3	0	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page